

Canteen Bill Calculator

JavaScript Mini Project | 1st Year BTech CSE | Time: 15-20 minutes

Topics used: var / let / const, normal functions, arrow functions, return values, console.log

Only JavaScript. No HTML, no CSS. Write your code in a .js file (or the browser console) and run it.

The Story

You have just been hired to build the **billing logic** for your college canteen. Right now the canteen uncle calculates every bill on paper, and it takes forever. Your job: write a small JavaScript program that takes the items a student bought and prints a clean, correct bill. You will use the same building blocks you learned in class: variables, functions that return values, and printing.

What You Will Practice

- Choosing between const and let correctly
- Writing a normal function and an arrow function
- Using return to give a value back, then printing that value
- Reading an error message (and not being scared of it!)

Your Tasks

Task 1: Set up the data

A bill needs some fixed information and some information that changes as items are added. Create variables for the following:

- The canteen name: "College Canteen" (it will never change)
- The GST rate: **5%**, stored as a decimal number (it will never change during billing)
- A **subtotal** that starts at **0** and grows as items are added

***Hint:** Ask yourself for each variable: "Will this value ever need to change later?" The answer decides the keyword. Also, 5% means 5 out of 100. How would you write that as a decimal?*

Task 2: Item total (use a NORMAL function)

Write a normal function called `itemTotal` that takes the **price** of one item and the **quantity** bought, and **returns** how much that item costs in total.

Example: 3 samosas at 15 each should give you 45.

***Hint:** Your function should hand the answer back, not just show it on the screen. Which keyword does that? Test it by calling the function and printing what comes back.*

Task 3: Add GST (use an ARROW function)

Write an arrow function called `addGST` that takes an **amount** and **returns** the amount after GST has been added.

Example: GST on 100 is 5, so the function should give back 105.

***Hint:** Work it out in two small steps on paper first: (1) how much is the GST on this amount? (2) what is the amount plus that GST? Then turn your two steps into one line of code. Use the GST variable you already created, don't type the rate again.*

Task 4: Student discount (use an ARROW function)

Write an arrow function called `applyDiscount` that takes an **amount** and a **percent** and **returns** the amount left after the discount is removed.

Example: 10% off 200 means you pay 180.

Hint: This function takes two inputs, so it needs two parameters. First find out how much money is being cut off, then take that away from the original amount. Remember, "percent" means "out of 100".

Task 5: Generate the bill

A student buys the items below. Use your functions to find each item's total.

Item	Price (per piece)	Quantity
Samosa	15	3
Tea	10	2
Sandwich	40	1

Add each item's total into `subtotal`, then use `console.log` to print your bill so that it looks **exactly** like this:

```
===== College Canteen =====  
Samosa total: 45  
Tea total: 20  
Sandwich total: 40  
Subtotal: 105  
Total with GST: 110.25  
=====
```

*Hint: Store each item's total in its own variable so you can print it and also add it up. To print text and a value on the same line, you can join them with the `+` sign. The first line of the bill uses your canteen name variable, so do not type the name again. Finally, check: does your Total with GST match **110.25**?*

Task 6: Break it on purpose!

Now be a detective. At the end of your program, try to **change the value** of one of your `const` variables (for example, make the GST rate 18%). Run the program and read the error message carefully.

Then, in a comment (a line starting with `//`), write **one sentence** explaining why the error happened.

Hint: Errors are not your enemy, they are JavaScript telling you exactly what went wrong. Look for the words in the message that describe the kind of problem and the name of the variable. Think back to what makes `const` different from `let`.

Bonus Challenges (for fast finishers)

- ★ **Bonus 1:** The canteen gives a **10% student discount**. Print the final amount after applying the discount *and* GST. Which should be applied first, the discount or the GST? Try both orders and compare.
- ★ **Bonus 2:** Add a **fourth item** of your choice (with your own price and quantity) and make sure the bill still adds up correctly.

Hint: You already wrote the discount function in Task 4. Think about how you can use the output of one function as the input of another.

Before You Submit: Self-Check

- ☐ Canteen name and GST rate use the correct keyword, and subtotal uses the other one
- ☐ `itemTotal` is a normal function and **returns** a value
- ☐ `addGST` and `applyDiscount` are arrow functions and **return** values
- ☐ My output matches the sample bill exactly, including **110.25**
- ☐ I tried changing a `const`, read the error, and wrote my explanation in a comment

Stuck? Re-read the hint, test one function at a time with `console.log`, and then ask your neighbour or your teacher. Debugging is a skill, and every programmer practises it daily.